Explaining acid rain

17a

Acid rain is a major problem in countries that are industrialised. Such countries tend to release large amounts of sulfur dioxide and nitrogen dioxide into the air. The former gas is emitted into the air by burning coal, which contains some sulfur, and by extracting metals from sulfide ores. The latter gas comes from the use of petrol and diesel in cars and trucks.

Nitrogen dioxide gas dissolves in rainwater drops to produce nitric acid. Sulfur dioxide in the atmosphere also dissolves in water droplets in the air to produce sulfuric acid. As a result of those acids in the rain drops, acid rain is formed.

Acid rain has a low pH, as low as 3. It is therefore quite acidic. The long-term effects of this rain have unfortunately lowered the pH of rivers and lakes. This has had a detrimental effect upon fish populations. Plants also suffer because the soil becomes affected as well. Such effects have been seen in North America and Scandinavia. In Europe, acid rain has done more damage to statues and old buildings than weathering has done in the last 400 years.

Looking at the text

1 Write down the meanings of the six underlined words in the text.

_____

_____

_____

_____

_____

_____

2 Write the main theme of each paragraph.

a Paragraph 1: _____

b Paragraph 2: _____

c Paragraph 3: _____

3 State the purpose of this text. _____

4 Pronouns are words such as *he, she, it* and *they*, which are used to refer to nouns that have been used earlier in the text. Find two pronouns in the text above and circle them. Then write down the pronouns and the noun they are referring to.

_____

_____

5 Another small but important part of text is the pointing words (also called determiners or articles). When you are asked to test for *a gas*, you may not know which gas you will find. But once you have finished testing, you know exactly which gas you are talking about, and from then on can refer to *the gas*. Therefore, using words like *a*, or no pointing word at all, keeps things very general, but using pointing words like *the* makes things very specific.

Find four specific pointing words in the text above and box them. Then write down the term with the pointing word and the specific term they are referring to.

_____

_____

_____

_____

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17b

Learning from the text

6 Suggest what effect acid rain would have on metals such as iron by writing word equations for the reactions. _____

7 Mortar usually contains limestone ($CaCO_3$). Write a word equation to show the effect acid rain would have mortar. _____

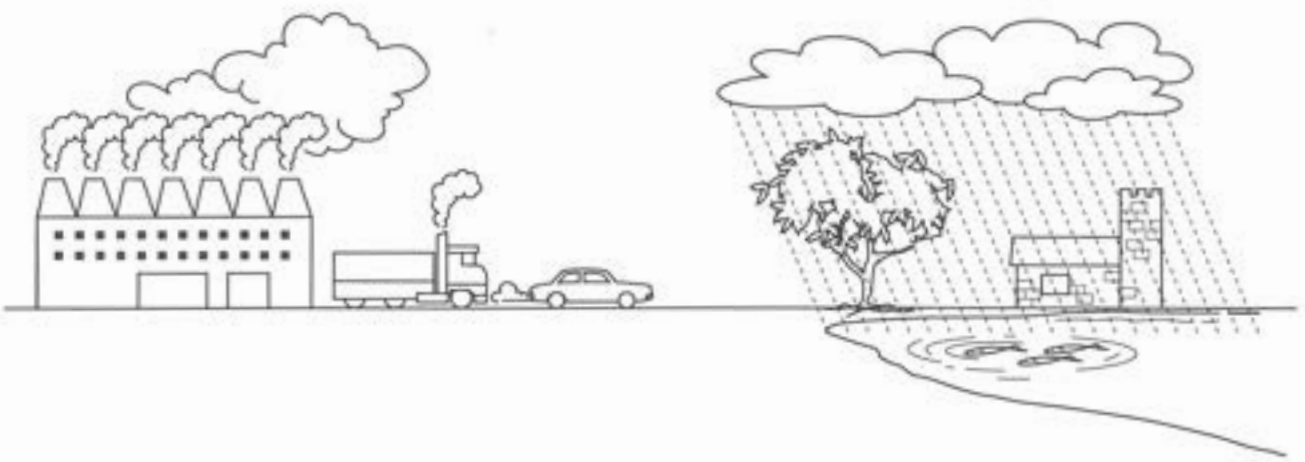
8 Explain why Australia is not strongly affected by acid rain. _____

_____

Labelling a diagram

9 Cut out the following labels and paste them in the appropriate places on the diagram below. Use the text above to help you.

1 Acids destroy trees and damage the soil.	2 Sulfur dioxide is emitted.	3 When nitrogen combusts in car engines, nitrogen dioxide is formed.
4 Sulfur dioxide and rain water combine to form sulfuric acid.	5 Nitrogen dioxide is given out.	6 When fossil fuels are burnt in factories, sulfur is changed into sulfur dioxide.
7 Combining nitrogen dioxide with rain water results in nitric acid.	8 Acids attack buildings and statues.	9 Acid rain lowers the pH of rivers and lakes and so reduces fish populations.



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